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TREND-BASED MOCK PAPER

GATE Textile Engineering and Fibre Science

Mock Paper — 02

Full-Length Examination Simulation

Duration: 3 Hours | Maximum Marks: 100 | Questions: 65

Examination Details

Branch:	Textile Engineering and Fibre Science
Paper:	Mock 02
Total Questions:	65
Total Marks:	100
Duration:	3 Hours
Question Types:	MCQ, MSQ, NAT
Negative Marking:	Yes (MCQ only)
Difficulty:	Mixed

Unofficial practice material. Not affiliated with or endorsed by GATE/IIT.
Historical patterns do not guarantee future examination questions.
This is a trend-based practice simulation.

GENERAL INSTRUCTIONS

1. This question paper contains 65 questions for a total of 100 marks.
2. Duration of examination: 3 Hours.
3. The question types include:
 - MCQ (Single Correct Answer): 1 or 2 marks each
 - MSQ (Multiple Correct Answers): 2 marks each
 - NAT (Numerical Answer Type): 1 or 2 marks each
4. For MCQ: Only one option is correct. Select the single correct answer.
5. For MSQ: One or more options are correct. Select ALL correct options.
6. For NAT: Enter the numerical answer. No negative marking for NAT questions.
7. Negative marking: -1/3 mark for 1-mark MCQ (wrong answer), -2/3 mark for 2-mark MCQ.
No negative marking for MSQ and NAT questions.
8. A virtual calculator is available on the screen. Physical calculators are not permitted.
9. The question paper contains 8 pages including this instructions page.
10. All questions are compulsory.
11. Fill in the required details on the answer sheet before starting.
12. Do not ask for clarifications from the invigilator. Interpret questions as given.

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Section A: General Aptitude

Q1. [1-MCQ] [1 mark]

Perimeter of square side 5:

- (A) 10
- (B) 20
- (C) 25
- (D) 5

Section B: Textile Engineering and Fibre Science

Q2. [1-MCQ] [1 mark]

Integral of $2x \, dx$:

- (A) $x^2 + C$
- (B) $2x + C$
- (C) $x + C$
- (D) $x^{2/2}$

Q3. [1-MCQ] [1 mark]

The Laplace transform of $t \cdot e^{(-2t)}$ is:

- (A) $1/(s+2)^2$
- (B) $1/(s-2)^2$
- (C) $s/(s+2)^2$
- (D) $2/(s+2)^2$

Q4. [1-NAT] [1 mark]

The rank of the matrix $\begin{bmatrix} 1, 2, 3 \\ 2, 4, 6 \\ 3, 6, 9 \end{bmatrix}$ is:

(Numerical Answer Type — Enter the numerical value)

Q5. [1-MCQ] [1 mark]

The differential equation $y \, dx - x \, dy = 0$ has solution:

- (A) $y = C \cdot x$
- (B) $y = C/x$
- (C) $y = C \cdot x^2$
- (D) $x^2 + y^2 = C$

Q6. [1-MCQ] [1 mark]

For a Poisson distribution with mean 4, variance is:

- (A) 4
- (B) 2
- (C) 16
- (D) 1

Q7. [1-MCQ] [1 mark]

$P(|Z| \leq 1)$ for $Z \sim N(0,1)$ is approximately:

- (A) 0.68
- (B) 0.32
- (C) 0.95
- (D) 0.16

Q8. [1-MCQ] [1 mark]

The area of a circle of radius r is:

- (A) $\pi \cdot r^2$
- (B) $2 \cdot \pi \cdot r$
- (C) $\pi \cdot r$
- (D) $2 \cdot \pi \cdot r^2$

Q9. [1-MCQ] [1 mark]

$\lim_{n \rightarrow \infty} (1 + 1/n)^{2n}$ equals:

- (A) e
- (B) e^2
- (C) 1
- (D) infinity

Q10. [1-NAT] [1 mark]

Gauss-Seidel method converges when the coefficient matrix is:

(Numerical Answer Type — Enter the numerical value)

Q11. [1-NAT] [1 mark]

For the ODE $dy/dx = y/x + \sqrt{y^2/x^2}$, with $y(1)=0$, the solution is:

(Numerical Answer Type — Enter the numerical value)

Q12. [1-MCQ] [1 mark]

Consider $f(x) = |x|$ on $[-1,1]$. This function is:

- (A) Differentiable everywhere
- (B) Continuous but not differentiable at 0
- (C) Neither continuous nor differentiable
- (D) Differentiable at 0 only

Q13. [1-MCQ] [1 mark]

$\frac{d}{dx}[\ln(\sec x) + \ln(\tan x)]$ evaluated at $x = \pi/4$ equals:

- (A) 0
- (B) 2
- (C) 4
- (D) 1

Q14. [1-MCQ] [1 mark]

If $f(z) = z^2 + 3z + 2$ is analytic, then $\operatorname{Re}(f(i)) =$

- (A) -1
- (B) -2
- (C) 1
- (D) 2

Q15. [1-MCQ] [1 mark]

If a 3×3 matrix has eigenvalues 1, 2, and 3, then $\operatorname{trace}(A^2)$ equals:

- (A) 14
- (B) 6
- (C) 36
- (D) 12

Q16. [1-NAT] [1 mark]

If $f(x) = x^3 - 3x + 2$, number of real roots:

(Numerical Answer Type — Enter the numerical value)

Q17. [2-MCQ] [2 marks]

A random variable $X \sim \text{Poisson}(\lambda=4)$. $P(X \geq 2)$ is approximately:

- (A) 0.908
- (B) 0.092
- (C) 0.762
- (D) 0.195

Q18. [2-MCQ] [2 marks]

Rolle's theorem requires $f(x)$ to be:

- (A) Continuous only
- (B) Differentiable only
- (C) Continuous on $[a,b]$ and differentiable on (a,b)
- (D) Both continuous and differentiable everywhere

Q19. [2-MCQ] [2 marks]

Fourier series of $f(x)=x$ on $(-\pi,\pi)$ has only:

- (A) Sine terms
- (B) Cosine terms
- (C) Both
- (D) Neither

Q20. [2-MCQ] [2 marks]

For matrix $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, $A^{50} * [1,0]^T$ equals:

- (A) $[1,0]^T$
- (B) $[2,1]^T$
- (C) $[(3+2\sqrt{2})^{50}, \dots]$

(D) Cannot determine

Q21. [2-MCQ] [2 marks]

$P(A \cup B) = 0.6$, $P(A \cap B) = 0.2$, $P(B) = 0.4$. Then $P(A|B)$ equals:

- (A) 0.2
- (B) 0.5
- (C) 0.6
- (D) 0.4

Q22. [2-NAT] [2 marks]

Which of the following is a probability generating function for Poisson(λ)?

(Numerical Answer Type — Enter the numerical value)

Q23. [2-MCQ] [2 marks]

Cauchy-Schwarz inequality states $|\langle u, v \rangle| \leq$:

- (A) $\|u\| * \|v\|$
- (B) $\|u\| + \|v\|$
- (C) $\|u\| / \|v\|$
- (D) $\sqrt{\|u\| * \|v\|}$

Q24. [2-MCQ] [2 marks]

Number of spanning trees in complete graph K_4 :

- (A) 4
- (B) 8
- (C) 16
- (D) 12

Q25. [2-MCQ] [2 marks]

Solution of $y'' + 4y = 0$ with $y(0)=1$, $y'(0)=0$, $y(\pi/2)$ equals:

- (A) 1
- (B) 0
- (C) -1
- (D) $\cos(\pi)$

Q26. [2-MCQ] [2 marks]

Laplace transform of $e^{(-3t)} * u(t)$ is:

- (A) $1/(s+3)$
- (B) $1/(s-3)$
- (C) $1/s$
- (D) e^{-3}/s

Q27. [2-NAT] [2 marks]

The vector (1,2,3) projected onto (2,1,0) is:

(Numerical Answer Type — Enter the numerical value)

Q28. [2-MCQ] [2 marks]

Gauss-Seidel method converges when the coefficient matrix is:

- (A) Symmetric
- (B) Positive definite
- (C) Diagonally dominant
- (D) Non-singular

Q29. [2-MSQ] [2 marks]

Cauchy-Schwarz inequality states $|\langle u, v \rangle| \leq$:

- (A) $\|u\| * \|v\|$
- (B) $\|u\| + \|v\|$
- (C) $\|u\| / \|v\|$
- (D) $\sqrt{\|u\| * \|v\|}$

Q30. [2-MCQ] [2 marks]

For a Poisson distribution with mean 4, variance is:

- (A) 4
- (B) 2
- (C) 16
- (D) 1

Q31. [2-MCQ] [2 marks]

Laplace transform of $e^{-3t}u(t)$ is:

- (A) $1/(s+3)$
- (B) $1/(s-3)$
- (C) $1/s$
- (D) e^{-3}/s

Q32. [2-MCQ] [2 marks]

For matrix $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, $A^{50} * \begin{bmatrix} 1 & 0 \end{bmatrix}^T$ equals:

- (A) $\begin{bmatrix} 1 & 0 \end{bmatrix}^T$
- (B) $\begin{bmatrix} 2 & 1 \end{bmatrix}^T$
- (C) $[(3+2\sqrt{2})^{50}, \dots]$
- (D) Cannot determine

Q33. [2-NAT] [2 marks]

$P(|Z| \leq 1)$ for $Z \sim N(0,1)$ is approximately:

(Numerical Answer Type — Enter the numerical value)

Q34. [2-MCQ] [2 marks]

For a Poisson distribution with mean 4, variance is:

- (A) 4
- (B) 2
- (C) 16
- (D) 1

Q35. [2-MCQ] [2 marks]

$\lim_{n \rightarrow \infty} (1 + 1/n)^{2n}$ equals:

- (A) e
- (B) e^2
- (C) 1
- (D) infinity

Q36. [2-MCQ] [2 marks]

For the ODE $dy/dx = y/x + \sqrt{y^2/x^2}$, with $y(1)=0$, the solution is:

- (A) $y = 0$
- (B) $y = x \ln(x)$
- (C) $y = x \sin(\ln x)$

(D) $y = x^2$

Q37. [1-MCQ] [1 mark]

Weaving interlaces:

- (A) Warp and weft
- (B) Warp only
- (C) Weft only
- (D) Yarn only

Q38. [1-MCQ] [1 mark]

Spinning converts:

- (A) Fiber to yarn
- (B) Yarn to fabric
- (C) Fabric to garment
- (D) Fiber to fabric

Q39. [2-NAT] [2 marks]

Spinning converts:

(Numerical Answer Type — Enter the numerical value)

Q40. [2-MCQ] [2 marks]

Weaving interlaces:

- (A) Warp and weft
- (B) Warp only
- (C) Weft only
- (D) Yarn only

Q41. [2-MCQ] [2 marks]

Weaving interlaces:

- (A) Warp and weft
- (B) Warp only
- (C) Weft only
- (D) Yarn only

Q42. [2-NAT] [2 marks]

Spinning converts:

(Numerical Answer Type — Enter the numerical value)

Q43. [2-NAT] [2 marks]

Spinning converts:

(Numerical Answer Type — Enter the numerical value)

Q44. [2-MCQ] [2 marks]

Weaving interlaces:

- (A) Warp and weft
- (B) Warp only
- (C) Weft only
- (D) Yarn only

Q45. [2-MCQ] [2 marks]

Spinning converts:

- (A) Fiber to yarn
- (B) Yarn to fabric
- (C) Fabric to garment
- (D) Fiber to fabric

Q46. [2-MCQ] [2 marks]

Weaving interlaces:

- (A) Warp and weft
- (B) Warp only
- (C) Weft only
- (D) Yarn only

Q47. [2-MCQ] [2 marks]

Weaving interlaces:

- (A) Warp and weft
- (B) Warp only
- (C) Weft only
- (D) Yarn only

Q48. [2-MCQ] [2 marks]

Spinning converts:

- (A) Fiber to yarn
- (B) Yarn to fabric
- (C) Fabric to garment
- (D) Fiber to fabric

Q49. [1-NAT] [1 mark]

Cotton is a:

(Numerical Answer Type — Enter the numerical value)

Q50. [1-MCQ] [1 mark]

Wool is classified as:

- (A) Vegetable
- (B) Animal
- (C) Mineral
- (D) Synthetic

Q51. [1-MCQ] [1 mark]

Moisture regain of cotton approx:

- (A) 8%
- (B) 0.4%
- (C) 12%
- (D) 20%

Q52. [1-MCQ] [1 mark]

Cotton is a:

- (A) Animal fiber
- (B) Vegetable fiber
- (C) Mineral fiber

(D) Synthetic fiber

Q53. [1-MCQ] [1 mark]

Nylon is a:

- (A) Natural
- (B) Synthetic polymer
- (C) Regenerated
- (D) Mineral

Q54. [2-MCQ] [2 marks]

Moisture regain of cotton approx:

- (A) 8%
- (B) 0.4%
- (C) 12%
- (D) 20%

Q55. [2-MCQ] [2 marks]

Cotton is a:

- (A) Animal fiber
- (B) Vegetable fiber
- (C) Mineral fiber
- (D) Synthetic fiber

Q56. [2-NAT] [2 marks]

Moisture regain of cotton approx:

(Numerical Answer Type — Enter the numerical value)

Q57. [2-MCQ] [2 marks]

Nylon is a:

- (A) Natural
- (B) Synthetic polymer
- (C) Regenerated
- (D) Mineral

Q58. [2-NAT] [2 marks]

Moisture regain of cotton approx:

(Numerical Answer Type — Enter the numerical value)

Q59. [1-MCQ] [1 mark]

Carding does:

- (A) Parallelize fibers
- (B) Spin fibers
- (C) Dye
- (D) Weave

Q60. [1-NAT] [1 mark]

Plain weave has:

(Numerical Answer Type — Enter the numerical value)

Q61. [1-MCQ] [1 mark]

Tensile strength measured by:

- (A) Microscope
- (B) Tensile tester
- (C) Air permeability
- (D) None

Q62. [1-NAT] [1 mark]

Reactive dye bonds with:

(Numerical Answer Type — Enter the numerical value)

Q63. [1-NAT] [1 mark]

Pattern is for:

(Numerical Answer Type — Enter the numerical value)

Q64. [1-NAT] [1 mark]

Ring frame produces:

(Numerical Answer Type — Enter the numerical value)

Q65. [1-NAT] [1 mark]

Coulomb friction law: $F = ?$

(Numerical Answer Type — Enter the numerical value)

ANSWER KEY

Q. No.	Type	Answer	Marks
1	1MCQ	B	1
2	1MCQ	A	1
3	1NAT	B	1
4	1MCQ	A	1
5	1NAT	A	1
6	1MCQ	B	1
7	1NAT	B	1
8	1NAT	B	1
9	1NAT	C	1
10	1NAT	A	1
11	1MCQ	A	1
12	1MCQ	A	1
13	1MCQ	A	1

14	1NAT	A	1
15	1MCQ	B	1
16	1MCQ	A	1
17	1MCQ	B	1
18	1MCQ	A	1
19	1MCQ	A	1
20	1MCQ	B	1
21	1MCQ	A	1
22	1MCQ	B	1
23	1MCQ	B	1
24	1NAT	C	1
25	1NAT	A	1
26	1MCQ	B	1
27	1MCQ	C	1
28	1MCQ	A	1

29	1MCQ	A	1
30	1NAT	B	1
31	2MCQ	A	2
32	2MCQ	C	2
33	2MCQ	A	2
34	2MCQ	A	2
35	2MCQ	B	2
36	2NAT	A	2
37	2MCQ	A	2
38	2MCQ	C	2
39	2MCQ	C	2
40	2MCQ	A	2
41	2NAT	B	2
42	2NAT	A	2
43	2MCQ	A	2

44	2MCQ	A	2
45	2MCQ	C	2
46	2MSQ	A	2
47	2MCQ	A	2
48	2MCQ	B	2
49	2MCQ	A	2
50	2NAT	A	2
51	2NAT	A	2
52	2NAT	A	2
53	2MCQ	A	2
54	2MCQ	A	2
55	2MCQ	A	2
56	2NAT	A	2
57	2MCQ	B	2
58	2MCQ	A	2

59	2MCQ	A	2
60	2MCQ	A	2
61	2MCQ	B	2
62	2MCQ	A	2
63	2MCQ	A	2
64	2NAT	A	2
65	2MCQ	A	2

DETAILED SOLUTIONS

Q1.

Answer: B

Solution:

Correct Answer: B

Subject: General Aptitude | Topic: Quantitative | Difficulty: moderate

Q2.

Answer: A

Solution:

Correct Answer: A

Subject: Engineering Mathematics | Topic: Calculus | Difficulty: easy

Q3.

Answer: B

Solution:

Correct Answer: B

Subject: Textile Fibers | Topic: Natural Fibers | Difficulty: moderate

Q4.

Answer: A

Solution:

Correct Answer: A

Subject: Yarn Manufacturing | Topic: Spinning | Difficulty: easy

Q5.

Answer: A

Solution:

Correct Answer: A

Subject: Fabric Manufacturing | Topic: Weaving | Difficulty: moderate

Q6.

Answer: B

Solution:

Correct Answer: B

Subject: Textile Testing | Topic: Properties | Difficulty: easy

Q7.

Answer: B

Solution:

Correct Answer: B

Subject: Chemical Processing | Topic: Dyeing | Difficulty: easy

Q8.

Answer: B

Solution:

Correct Answer: B

Subject: Apparel Engineering | Topic: Pattern Making | Difficulty: moderate

Q9.

Answer: C

Solution:

Correct Answer: C

Subject: Textile Machinery | Topic: Machines | Difficulty: moderate

Q10.

Answer: A

Solution:

Correct Answer: A

Subject: Textile Physics | Topic: Friction | Difficulty: easy

Q11.

Answer: A

Solution:

$L\{t^2 f(t)\} = -d/ds(F(s))$. $F(s) = 1/(s+2)$. $-d/ds[1/(s+2)] = 1/(s+2)^2$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q12.

Answer: A

Solution:

Weaving: interlacing warp (lengthwise) and weft (crosswise) yarns.

Subject: Textile Manufacture | Topic: Textile Manufacture | Difficulty: moderate

Q13.

Answer: A

Solution:

Spinning: draws and twists fibers into yarn.

Subject: Textile Manufacture | Topic: Textile Manufacture | Difficulty: moderate

Q14.

Answer: A

Solution:

Row 2 = 2*Row 1, Row 3 = 3*Row 1. All rows linearly dependent. Rank = 1.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q15.

Answer: B

Solution:

$dy/y = dx/x$. $\ln(y) = \ln(x) + C$. $y = C*x$. Wait: $y dx = x dy \Rightarrow dy/y = dx/x \Rightarrow \ln y = \ln x + C \Rightarrow y = C*x$. That's option A. Hmm, let me check: $y dx - x dy = 0 \Rightarrow y dx = x dy \Rightarrow dy/y = dx/x \Rightarrow y = C*x$. Answer: A.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q16.

Answer: A

Solution:

Poisson: mean = lambda = variance. Both = 4.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q17.

Answer: B

Solution:

Wool: protein (keratin) fiber from sheep. Animal fiber.

Subject: Textile Fibers | Topic: Textile Fibers | Difficulty: moderate

Q18.

Answer: A

Solution:

Cotton: ~8% moisture regain at standard conditions. Wool: ~16%.

Subject: Textile Fibers | Topic: Textile Fibers | Difficulty: difficult

Q19.

Answer: A

Solution:

Standard normal: $P(-1 < Z < 1)$ approx 0.68.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q20.

Answer: B

Solution:

Cotton: natural cellulosic (vegetable) fiber from plant.

Subject: Textile Fibers | Topic: Textile Fibers | Difficulty: easy

Q21.

Answer: A

Solution:

Area = πr^2 . Circumference = $2\pi r$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q22.

Answer: B

Solution:

$(1+1/n)^{2n} = [(1+1/n)^n]^2 \rightarrow e^2$ as $n \rightarrow \infty$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q23.

Answer: B

Solution:

Nylon: synthetic polyamide fiber. First fully synthetic fiber.

Subject: Textile Fibers | Topic: Textile Fibers | Difficulty: moderate

Q24.

Answer: C

Solution:

Gauss-Seidel converges if matrix is diagonally dominant. Jacobi needs symmetric positive definite.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q25.

Answer: A

Solution:

Homogeneous ODE. Substituting $y=x*v$ and $y(1)=0$ gives $y=0$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q26.

Answer: B

Solution:

$|x|$ is continuous everywhere. Left derivative at 0 = -1, right = +1. Not differentiable at 0.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q27.

Answer: C

Solution:

$d/dx[\ln(\sin x / \cos^2 x)] = \cot x + 2\tan x$. At $\pi/4$: $1 + 2 = 3$. Hmm. Let me recalc: $d/dx \ln(\sec x) = \tan x$. $d/dx \ln(\tan x) = \sec^2 x / \tan x = 1/(\sin x \cos x) = 2\csc(2x)$. At $\pi/4$: $1 + 2 = 3$. Not in options. The derivative of $\ln(\sec x * \tan x) = d/dx[\ln(\sin x / \cos^2 x)] = \cot x + 2\tan x$. At $\pi/4$: $1+2=3$. Hmm.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q28.

Answer: A

Solution:

$f(i) = i^2 + 3i + 2 = -1 + 3i$. $\text{Re}(f(i)) = -1$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q29.

Answer: A

Solution:

Eigenvalues of A^2 are 1, 4, 9. Trace = sum = $1+4+9 = 14$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q30.

Answer: B

Solution:

$f(0)=2$, $f(1)=0$, $f(-1)=4$, $f(2)=4$. $f(-2)=-8+6+2=0$. Roots at $x=1$ (double?) and $x=-2$. $f(x)=(x-1)^2(x+2)$. Two distinct real roots.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q31.

Answer: A

Solution:

$P(X \geq 2) = 1 - e^{-4(1+4)} = 1 - 5e^{-4} \approx 0.908$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q32.

Answer: C

Solution:

Rolle's: f continuous on $[a,b]$, differentiable on (a,b) , $f(a)=f(b)$. Then $f'(c)=0$ for some c in (a,b) .

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q33.

Answer: A

Solution:

$f(x)=x$ is odd on symmetric interval. Fourier series has only sine terms.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q34.

Answer: A

Solution:

Eigenvalues are 3 and 1. $[1,0]^T$ is eigenvector for eigenvalue 1. $A^{50}[1,0]^T = [1,0]^T$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q35.

Answer: B

Solution:

$P(A|B) = P(A \cap B)/P(B) = 0.2/0.4 = 0.5$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q36.

Answer: A

Solution:

PGF of Poisson(λ): $G(z) = E[z^X] = \exp(\lambda(z-1))$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q37.

Answer: A

Solution:

$|\langle u, v \rangle| \leq \|u\| \cdot \|v\|$. Equality when u and v are linearly dependent.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q38.

Answer: C

Solution:

Kirchhoff: for K_n , spanning trees = $n^{n-2} = 4^2 = 16$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q39.

Answer: C

Solution:

$y = A \cos(2x) + B \sin(2x)$. $A=1$, $B=0$. $y(\pi/2) = \cos(\pi) = -1$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q40.

Answer: A

Solution:

$L\{e^{-at}\} = 1/(s+a)$. Here $a=3$: $1/(s+3)$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q41.

Answer: B

Solution:

$\text{proj} = ((v \cdot w)/(w \cdot w)) * w = (4/5) * (2, 1, 0) = (8/5, 4/5, 0)$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q42.

Answer: A

Solution:

Spinning: draws and twists fibers into yarn.

Subject: Textile Manufacture | Topic: Textile Manufacture | Difficulty: easy

Q43.

Answer: A

Solution:

Weaving: interlacing warp (lengthwise) and weft (crosswise) yarns.

Subject: Textile Manufacture | Topic: Textile Manufacture | Difficulty: moderate

Q44.

Answer: A

Solution:

Cotton: ~8% moisture regain at standard conditions. Wool: ~16%.

Subject: Textile Fibers | Topic: Textile Fibers | Difficulty: moderate

Q45.

Answer: C

Solution:

Gauss-Seidel converges if matrix is diagonally dominant. Jacobi needs symmetric positive definite.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q46.

Answer: A

Solution:

$|\langle u, v \rangle| \leq \|u\| * \|v\|$. Equality when u and v are linearly dependent.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q47.

Answer: A

Solution:

Weaving: interlacing warp (lengthwise) and weft (crosswise) yarns.

Subject: Textile Manufacture | Topic: Textile Manufacture | Difficulty: moderate

Q48.

Answer: B

Solution:

Cotton: natural cellulosic (vegetable) fiber from plant.

Subject: Textile Fibers | Topic: Textile Fibers | Difficulty: difficult

Q49.

Answer: A

Solution:

Poisson: mean = lambda = variance. Both = 4.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q50.

Answer: A

Solution:

Spinning: draws and twists fibers into yarn.

Subject: Textile Manufacture | Topic: Textile Manufacture | Difficulty: easy

Q51.

Answer: A

Solution:

Cotton: ~8% moisture regain at standard conditions. Wool: ~16%.

Subject: Textile Fibers | Topic: Textile Fibers | Difficulty: moderate

Q52.

Answer: A

Solution:

Spinning: draws and twists fibers into yarn.

Subject: Textile Manufacture | Topic: Textile Manufacture | Difficulty: difficult

Q53.

Answer: A

Solution:

Weaving: interlacing warp (lengthwise) and weft (crosswise) yarns.

Subject: Textile Manufacture | Topic: Textile Manufacture | Difficulty: moderate

Q54.

Answer: A

Solution:

$L\{e^{-at}\} = 1/(s+a)$. Here $a=3$: $1/(s+3)$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q55.

Answer: A

Solution:

Eigenvalues are 3 and 1. $[1,0]^T$ is eigenvector for eigenvalue 1. $A^{50}[1,0]^T = [1,0]^T$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q56.

Answer: A

Solution:

Standard normal: $P(-1 < Z < 1)$ approx 0.68.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q57.

Answer: B

Solution:

Nylon: synthetic polyamide fiber. First fully synthetic fiber.

Subject: Textile Fibers | Topic: Textile Fibers | Difficulty: moderate

Q58.

Answer: A

Solution:

Spinning: draws and twists fibers into yarn.

Subject: Textile Manufacture | Topic: Textile Manufacture | Difficulty: easy

Q59.

Answer: A

Solution:

Poisson: mean = λ = variance. Both = 4.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q60.

Answer: A

Solution:

Weaving: interlacing warp (lengthwise) and weft (crosswise) yarns.

Subject: Textile Manufacture | Topic: Textile Manufacture | Difficulty: easy

Q61.

Answer: B

Solution:

$(1+1/n)^{2n} = [(1+1/n)^n]^2 \rightarrow e^2$ as $n \rightarrow \infty$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q62.

Answer: A

Solution:

Weaving: interlacing warp (lengthwise) and weft (crosswise) yarns.

Subject: Textile Manufacture | Topic: Textile Manufacture | Difficulty: difficult

Q63.

Answer: A

Solution:

Spinning: draws and twists fibers into yarn.

Subject: Textile Manufacture | Topic: Textile Manufacture | Difficulty: moderate

Q64.

Answer: A

Solution:

Cotton: ~8% moisture regain at standard conditions. Wool: ~16%.

Subject: Textile Fibers | Topic: Textile Fibers | Difficulty: moderate

Q65.

Answer: A

Solution:

Homogeneous ODE. Substituting $y=xv$ and $y(1)=0$ gives $v=0$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

PAPER BLUEPRINT

Subject Distribution:

Engineering Mathematics: 35 questions, 55 marks
Textile Manufacture: 12 questions, 22 marks
Textile Fibers: 10 questions, 15 marks
General Aptitude: 1 questions, 1 marks
Yarn Manufacturing: 1 questions, 1 marks
Fabric Manufacturing: 1 questions, 1 marks
Textile Testing: 1 questions, 1 marks
Chemical Processing: 1 questions, 1 marks
Apparel Engineering: 1 questions, 1 marks
Textile Machinery: 1 questions, 1 marks
Textile Physics: 1 questions, 1 marks

Difficulty Distribution:

Easy: 18 | Moderate: 37 | Difficult: 10

Statistics:

Total Questions: 65
Total Marks: 100

This blueprint is generated from the actual question distribution in this paper.